

**Calibration Under Changing Market Conditions:  
Evaluating the Robustness of the Heston Stochastic  
Volatility Model**

Justine Odegbami

Quantitative Finance and Data Science MSc

Supervisor: Dr Simon Hubbert

September 2026

## Abstract

The Heston model is a widely adopted stochastic volatility model used in the pricing and hedging of options. However, it has a semi-closed form solution, meaning that parameter estimation requires numerical calibration, and the choice of optimiser can materially affect the resulting fit. There is added complexity in the context of stressed market regimes, wherein the model must accommodate a less smooth and more rapidly changing volatility surface. This is under-explored in the literature, and so this paper investigates the question: to what extent does the chosen method of calibration affect the accuracy and stability of the Heston model when the underlying market regime changes from calm to stressed?

On evaluation of methodologies investigated in the literature, this paper uses COS as the pricing engine, equally-weighted implied volatility RMSE as the loss function, and compares three calibration approaches:

- Levenberg-Marquardt (LM), a gradient-based local optimiser
- Nelder-Mead (NM), a derivative-free simplex method
- Differential Evolution (DE), a global population-based optimiser

These are applied to SPX options data covering a calm period (1st January - 31st October 2021) and a sustained stress period (1st January - 31st October 2022), with ten calibration dates per regime. In-sample calibration error, parameter stability across dates, out-of-sample pricing performance, and computational cost are each measured, alongside visual comparison of fitted vs market implied volatility surfaces for a representative date in each regime.

Results find that LM and DE consistently converged to near-identical parameter estimates and in-sample fit, despite using fundamentally different search mechanisms, which incidentally suggests a well-defined and findable optimum across both regimes. NM produced a meaningfully weaker in-sample fit throughout, with greater variability across dates. Regarding parameter stability, LM produced the most consistent estimates in the calm regime, while all three methods were broadly comparable under stress. Out-of-sample degradation was similar across all three optimisers in the stressed regime. Computationally, LM required approximately 45 times fewer function evaluations than DE while achieving essentially equiva-

lent accuracy. These findings have direct implications for practitioners choosing calibration methods in trading, risk, or model validation contexts.

# Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                             | <b>6</b>  |
| 1.1      | Motivation . . . . .                                            | 6         |
| 1.2      | Objectives . . . . .                                            | 8         |
| <b>2</b> | <b>Literature Review</b>                                        | <b>9</b>  |
| 2.1      | Stochastic Volatility Models and the Heston Framework . . . . . | 10        |
| 2.1.1    | The Heston Equations . . . . .                                  | 10        |
| 2.1.2    | The Feller Condition . . . . .                                  | 12        |
| 2.1.3    | Gil-Pelaez Inversion and Price Recovery . . . . .               | 12        |
| 2.1.4    | Empirical Performance and Known Limitations . . . . .           | 14        |
| 2.2      | COS Inversion and Prior Numerical Methods . . . . .             | 15        |
| 2.3      | Calibration Methodology . . . . .                               | 19        |
| 2.3.1    | Calibration Approach and Loss Function . . . . .                | 19        |
| 2.3.2    | Optimisation Methods . . . . .                                  | 20        |
| 2.4      | Calibration During Stressed Regimes . . . . .                   | 22        |
| <b>3</b> | <b>Methodology</b>                                              | <b>24</b> |
| 3.1      | Research Design Overview . . . . .                              | 24        |
| 3.2      | Data . . . . .                                                  | 24        |
| 3.3      | Pricing Implementation . . . . .                                | 25        |
| 3.4      | Loss Function . . . . .                                         | 26        |
| 3.5      | Starting Parameters . . . . .                                   | 26        |
| 3.6      | Optimiser Configuration . . . . .                               | 26        |
| 3.7      | Validity Screening . . . . .                                    | 27        |
| 3.8      | Post-Analysis . . . . .                                         | 27        |
| <b>4</b> | <b>Results</b>                                                  | <b>29</b> |
| 4.1      | Parameter Stability . . . . .                                   | 30        |
| 4.2      | Computational Cost . . . . .                                    | 31        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| 4.3      | Out-of-Sample Performance . . . . .                        | 32        |
| 4.4      | Implied Volatility Surfaces . . . . .                      | 33        |
| <b>5</b> | <b>Discussion</b>                                          | <b>36</b> |
| 5.1      | Does Optimiser Choice Matter? . . . . .                    | 36        |
| 5.2      | Parameter Instability: Procedural or Structural? . . . . . | 37        |
| 5.3      | Limitations . . . . .                                      | 37        |
| <b>6</b> | <b>Conclusion</b>                                          | <b>39</b> |
| 6.1      | Practical Implications . . . . .                           | 39        |
| 6.2      | Future Work . . . . .                                      | 39        |
| <b>7</b> | <b>Appendix</b>                                            | <b>41</b> |
| 7.1      | DE Raw Data . . . . .                                      | 41        |
| 7.2      | LM Raw Data . . . . .                                      | 42        |
| 7.3      | NM Raw Data . . . . .                                      | 43        |

# 1 Introduction

## 1.1 Motivation

Derivatives markets require accurate and timely pricing, of options and otherwise. A variety of pricing methods and formulae are available, within which volatility modelling is approached in many different ways.

Black, Scholes & Merton (1973) (BSM) is a foundational pricing formula, which provides a tractable closed-form solution and so is lauded for its relative ease of use. However, it makes the fundamental assumption of a constant volatility, which fails to reproduce the well-documented volatility smile present in real-world market observation. Very soon after its publication, Merton introduced jump diffusion (Merton 1976), arguably an implicit acknowledgement that the continuous diffusion model of Black-Scholes is not a sufficient representation of reality. Jump diffusion better accommodates sudden shocks - as became a reality on October 19th 1987.

On what has since been dubbed "Black Monday", equity markets worldwide lost between a fifth and two-thirds of their value in a matter of hours. Prior to this, the flat volatility surface implied by BSM had been sufficient; a world in which a 60% drop in markets' value in one day was a mathematical near-impossibility. But suddenly, it was no longer just academic; investors urgently needed the reassurance of a model that properly accommodated extreme behaviour. After Black Monday, the implied volatility skew (sometimes dubbed a "smirk") became structural and persistent as deep out-of-the-money put options (which pay out only during severe market declines) began trading at systematically elevated prices as market participants priced in the now non-negligible possibility of another similar catastrophe.

Hull & White (1987), Scott (1987), Wiggins (1987), and Stein & Stein (1991) produced the first wave of stochastic volatility models, and it is no coincidence that they emerged right after the crash. Unfortunately but unsurprisingly, these early models were essentially analytically intractable as they had no closed-form solution and so required repeated Monte Carlo for calibration, rendering them computationally prohibitive.

The breakthrough made in Heston (1993) was the derivation of a characteristic function, which leads to a semi-closed solution - a huge improvement in tractability over earlier stochastic volatility models without compromising on the better smile fit stochastic volatility models provide. The main innovation in stochastic models is the volatility-of-variance parameter which, as it sounds, models the volatility of the variance term. If this is set to zero, the stochastic component of the model collapses and reduces it exactly to BSM. Otherwise, for a positive volatility-of-variance, Heston generates a richer return distribution that prices out-the-money options above and in-the-money options below the equivalent BSM price at the same volatility levels, while the two models converge almost exactly at-the-money. This cemented Heston as the prevailing stochastic volatility model.

In practice, a calibrated Heston model serves several downstream purposes. Where liquid market prices are unavailable - for exotic options, barrier products, or strikes and maturities between quoted points - the calibrated model provides a theoretically consistent mechanism for interpolating the implied volatility surface and pricing contracts for which no direct market quote exists. The calibrated parameter set also determines the model's Greeks, which practitioners use to construct delta hedges and measure sensitivity to underlying risk factors. The quality of all such downstream outputs is therefore contingent on the quality of the calibration.

Calibration begins by inferring starting points for the five Heston parameters from market prices. Since the parameters are unobservable, they cannot be directly estimated, so instead a numerical optimiser searches the parameter space for the combination that minimises the distance between the implied volatility surface created by the model and the current market surface. The pricing engine throughout this study is the characteristic function derived by Heston (1993). It is inverted via the COS method of Fang & Oosterlee (2008) which converts a candidate parameter set into a set of prices for all quoted strikes and maturities. These are then compared to market prices via a loss function, which in this study is the (equally-weighted) implied volatility RMSE. The optimiser iteratively proposes improved parameter sets until a convergence tolerance is met. It is this calibration loop - and specifically the choice of optimiser driving it - that forms the

subject of this investigation.

## 1.2 Objectives

This paper investigates if the choice of calibration optimiser can affect Heston model accuracy and parameter stability across calm and stressed market regimes. The findings are relevant to model risk management and quant desk calibration design.

This study achieves this with a controlled comparison of local and global Heston calibration optimisers applied to both calm and stressed market regimes - an area the existing literature (which predominantly uses calm market data) has left largely unexplored. In doing so, it provides a quantitative basis for assessing the extent to which Heston's degradation in stressed markets is attributable to calibration methodology as opposed to fundamental model limitations.

Section 2 reviews the literature, covering the Heston framework, numerical inversion methods, calibration methodology, and various optimisation approaches, with key mathematical results derived throughout. In light of the conclusions drawn from Section 2, Section 3 describes the data and methodology adopted for the study, including data sourcing and filtering, pricing implementation, optimiser configuration, and post-analysis investigation. Section 4 presents the calibration results across both regimes and all three optimisers, alongside post-analysis results. Section 5 interprets these findings and discusses their practical implications. Section 6 concludes with a summary of the findings, the study's contributions, and directions for future work. Find the raw output of every model calibration run in the appendix.

## 2 Literature Review

The BSM model (Black et al. 1973) provides a closed-form solution for European option prices under the assumption that asset returns follow a geometric Brownian motion with constant drift and constant volatility. Its widespread adoption in practice owes as much to analytical tractability as to empirical quality: pricing requires only a single unknown parameter (implied volatility) which can be backed out from any observed option price by relatively simple numerical inversion, and reduces calibration to a straight-forward root-finding problem. This parsimony naturally made BSM the benchmark against which all subsequent pricing models are measured.

The BSM formula prices a European call as:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \quad (1)$$

where  $S_0$  is the underlying price,  $N(\cdot)$  is the standard normal cumulative distribution function,  $K$  is the strike price,  $r$  is the risk free rate,  $T$  is the time to maturity and

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \quad (2)$$

where  $\sigma$  is the (constant) volatility. Implied volatility (i.e. the value of volatility that would lead to an option price matching market price) is attained by backing this out from an observed option price via numerical inversion.

The model's central assumption of constant volatility, however, can be easily demonstrated as incompatible with observed market prices. When the BSM formula is backed out for options sharing the same underlying and expiry but different strike, the implied volatilities obtained are not constant as necessitated by the model. Rubinstein (1985) detailed this relationship - dubbed the volatility "smile" due to its parabolic shape - and showed that both the associated pricing biases are time-dependent. The best example of this is the pronounced asymmetric skew that emerged from and has persisted since the market crash of October 1987, which cannot be accommodated by BSM's thin-tailed

log-normal return distribution.

The same inconsistency exists across the maturity dimension. When BSM is backed out for options sharing the same strike but different expiry, implied volatilities again vary, producing a term structure for implied volatility that BSM's fundamental assumption of constant volatility is unable to reconcile. Together, the smile and the term structure define the implied volatility surface: a three-dimensional map of BSM's attempt at option pricing across all traded strikes and maturities simultaneously. Bakshi, Cao & Chen (1997) establish formally that BSM exhibits systematic pricing biases across both dimensions, and that these biases are structural rather than random. Any model that seeks to improve on BSM must therefore be capable of reproducing the shape of this surface from a parsimonious set of interpretable parameters.

## **2.1 Stochastic Volatility Models and the Heston Framework**

### **2.1.1 The Heston Equations**

The inadequacy of a constant volatility when applied to a post-crash market motivated a first wave of stochastic volatility models in which the variance is itself driven by a separate stochastic process. Hull & White (1987), Scott (1987), and Wiggins (1987) each proposed early specifications of stochastic volatility models which had marked improvements over BSM in fitting observed option prices. However, these models shared a fundamental limitation: the absence of closed-form pricing solutions required computationally intensive numerical methods to solve two-dimensional partial differential equations, thus making large-scale calibration essentially impossible. Stein & Stein (1991) subsequently proposed a more analytically tractable version, but only achieved this by assuming zero correlation between the volatility process and the underlying asset... a somewhat unrealistic restriction that prevents the model from generating the skewness necessary to reproduce the post-1987 implied volatility skew present in equity markets.

Heston (1993) resolved both limitations simultaneously. The Heston model specifies an asset price process in which the instantaneous variance follows a mean-reverting

square-root Cox-Ingersoll-Ross type (Cox, Ingersoll & Ross 1985) of diffusion:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S \quad (3)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^v \quad (4)$$

where  $S_t$  is the asset price,  $\mu$  is the drift,  $v_t$  is the instantaneous variance, and  $dW_t^S$  and  $dW_t^v$  are correlated Wiener increments with:

$$dW_t^S dW_t^v = \rho dt \quad (5)$$

The five remaining parameters together determine the shape of the implied volatility surface. Cui, del Baño Rollin & Germano (2017) characterise the role of each:

- Mean reversion speed  $\kappa \geq 0$ : controls the curvature with  $\sigma$ ; determines how quickly variance reverts to  $\theta$
- Long-run variance  $\theta \geq 0$ : drives term structure; the level to which variance mean-reverts
- Volatility of variance  $\sigma \geq 0$ : controls the curvature of the smile with  $\kappa$ ; models the volatility of the asset variance
- Correlation  $\rho \in [-1, 1]$ : typically negative in equity markets as falling prices tend to coincide with rising volatility; controls the skewness of the smile
- Initial variance  $v_0 \geq 0$ : governs the overall level of the smile; decides the starting value for variance

Over long horizons, spot returns become asymptotically normal as the variance process mean-reverts to  $\theta$ , implying that BSM recovers its accuracy for very long-dated options and so Heston's improvements over BSM are concentrated at short-to-medium maturities (Heston 1993).

### 2.1.2 The Feller Condition

The Feller condition  $2\kappa\theta > \sigma^2$  ensures the variance  $v_t$  remains strictly positive. This matters because if  $v_t$  reaches zero, the diffusion term  $\sigma\sqrt{v_t}$  vanishes entirely, meaning the variance process is momentarily deterministic. At that point, the asset price loses its stochastic component and the implied volatility surface collapses. The Feller condition ensures this cannot happen by guaranteeing the mean-reverting drift is always strong enough to keep  $v_t$  strictly above zero. As  $v_t$  approaches zero, the diffusion term  $\sigma\sqrt{v_t}$  vanishes while the "drift" term  $\kappa(\theta - v_t) = \kappa\theta$  stays positive, meaning at that point, the process is being pushed away from zero by its drift. Cox et al. (1985) show that this mean-reverting drift is enough to prevent the process from ever reaching zero only if  $2\kappa\theta > \sigma^2$ , as has been formalised.

However, Cui et al. (2017) note that calibrated parameters frequently violate this condition in practice without causing numerical instability, and that relaxing it is not only harmless but giving the calibrator freedom is often necessary for a good surface fit. This is because despite the violation, the characteristic function can remain well-behaved provided the parameters are economically reasonable.

### 2.1.3 Gil-Pelaez Inversion and Price Recovery

Heston's primary contribution is a semi-closed-form pricing formula derived via the characteristic function of the log-asset price distribution. While the density  $f(y|x)$  of the log-asset price is not available in closed form for the Heston model, what is available is its Fourier transform i.e. the characteristic function  $\phi_j$ . The Fourier transform of a function  $f(x)$  is defined as

$$\hat{f}(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx \quad (6)$$

meaning it decomposes  $f(x)$  into its frequency components across all frequencies  $u$ , for a real variable  $x$ . The Fourier transform of a probability density yields the characteristic function of the log-asset price, and then Fourier inversion recovers the risk-neutral probabilities  $P1$  and  $P2$  needed to price the European call. Following the same structure as

BSM, the European call price is expressed as:

$$C = SP_1 - Ke^{-r(T-t)}P_2 \quad (7)$$

where  $P_2$  is the probability under the risk-free measure that the option expires in the money, and  $P_1$  is the analogous probability under the stock price measure, which discounts using the underlying price rather than the risk-free rate. Neither  $P_1$  nor  $P_2$  is available in closed form directly.

The characteristic function  $\phi_j$  is defined as:

$$\phi_j(u) = E_{\mathbb{Q}_j}[\exp(iu \ln S_T)], \quad j = 1, 2 \quad (8)$$

where  $\mathbb{Q}_1$  and  $\mathbb{Q}_2$  denote the underlying stock price and risk-free measures respectively. Heston (1993) showed that  $\phi_1$  and  $\phi_2$  satisfy the same partial differential equations as bond prices in the Cox-Ingersoll-Ross model, and derived explicit closed-form expressions for both. They have the form:

$$\phi_j(u; \tau) = \exp(C_j(u, T-t) + D_j(u, T-t)v_0 + iu \ln S_0), \quad j = 1, 2 \quad (9)$$

where  $T-t$  is the time to maturity, and:

$$\begin{aligned} u_1 &= \frac{1}{2}, & u_2 &= -\frac{1}{2} \\ b_1 &= \kappa + \lambda - \rho\sigma, & b_2 &= \kappa + \lambda \\ d_j &= \sqrt{(\rho\sigma iu - b_j)^2 - \sigma^2(2u_j iu - u^2)} \\ g_j &= \frac{b_j - \rho\sigma iu + d_j}{b_j - \rho\sigma iu - d_j} \\ D_j &= \frac{b_j - \rho\sigma iu + d_j}{\sigma^2} \frac{1 - \exp(d_j\tau)}{1 - g_j \exp(d_j\tau)} \\ C_j &= r\tau iu + \frac{\kappa\theta}{\sigma^2} \left[ (b_j - \rho\sigma iu + d_j)\tau - 2 \ln \frac{1 - g_j \exp(d_j\tau)}{1 - g_j} \right] \end{aligned}$$

where  $\lambda$  is a market risk premium parameter that can be trivially set to 0. The function

differs between  $j = 1$  and  $2$  only through  $u_j$  and  $b_j$ . The probabilities are then recovered via:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[ \frac{\exp(-iu \ln K) \phi_j(u)}{iu} \right] du, \quad j = 1, 2 \quad (10)$$

Gil-Pelaez (1951) showed that the cumulative probability  $P_j$  can be expressed directly in terms of  $\phi_j$  via Fourier inversion without recovering the density itself. Heston (1993) applied this result to recover the probabilities  $P_1$  and  $P_2$ , yielding equation (10). The integrand decays rapidly and smoothly, making numerical evaluation efficient. This approach (which Heston notes can be extended to FX and even bond options) irrefutably demonstrates the practical advantage Heston has over all prior stochastic volatility models and is why it remains the industry standard.

#### 2.1.4 Empirical Performance and Known Limitations

The empirical performance of multiple types of stochastic volatility models relative to BSM was assessed systematically by Bakshi et al. (1997), who compared BSM against pure stochastic volatility (SV), SV with jumps (SVJ), SV with stochastic interest rates (SVSI), and a method that combines them all (SVSIJ). The jump component is based on the framework of Merton (1976), who introduced a jump-diffusion specification to accommodate the sudden, large price movements that occur with important new information arrivals. They conclude a clear order of contribution priority:

1. stochastic volatility is the single most important improvement over BSM
2. adding jumps further reduces pricing errors, especially at short maturities where variance alone cannot swing enough to produce the necessary kurtosis without an uneconomically large vol-of-vol
3. stochastic interest rates primarily improve the discounting of payoffs on long-dated contracts
4. neither extension sufficiently improves hedging performance over SV alone

A further limitation of the Heston framework concerns the dynamics of volatility itself. The roughness of a random path can be measured by its Hurst exponent  $H$ , a

value between 0 and 1, where  $H = 0.5$  corresponds to a standard Brownian motion, while values below 0.5 indicate a "rougher", more erratic path. Gatheral, Jaisson & Rosenbaum (2018) estimate that real-world volatility exhibits  $H \approx 0.1$  - much rougher than the  $H = 0.5$  Brownian motion of the Heston variance process (equation (4)). This suggests Heston may be structurally unable to capture the behaviour of realised volatility.

Christoffersen, Heston & Jacobs (2009) further find that Heston's single-factor volatility specification exhibits parameter instability across maturities and market conditions, while multi-factor stochastic volatility models provide a more stable representation. This possibly highlights a broader limitation of the Heston framework: a good surface fit does not necessarily imply that its parameters provide a stable description of volatility dynamics across different market environments.

## 2.2 COS Inversion and Prior Numerical Methods

Heston calibration requires the pricing formula to be evaluated thousands of times as the optimiser traverses the parameter space, so the computational cost of Fourier inversion directly decides calibration feasibility. Carr & Madan (1999) proposed the Fast Fourier Transform (FFT) as an efficient way to do this. FFT expresses the option price as the Fourier transform of a damped log-strike payoff function and it recovers option price  $C$  for a grid of strikes in a single  $O(N \log N)$  transform.

$$C(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \exp(-iuk) \psi(u) du \quad (11)$$

where  $k = \ln K$  is the log-strike price,  $\alpha > 0$  is the damping parameter, and:

$$\psi(u) = \frac{e^{-rT} \phi(u - (\alpha + 1)i)}{\alpha^2 + \alpha - u^2 + i(2\alpha + 1)u} \quad (12)$$

where  $u$  are Fourier domain frequencies,  $\phi$  is the characteristic function of the log-asset price, and  $\psi$  is known as the Fourier transform of the modified call price. The damping is necessary because the log-strike of call payoffs is not square-integrable, which violates the integrability condition imposed by Fourier inversion.

A further complication arises from the complex component of the Heston characteristic function. Lord & Kahl (2006) identify that implementation introduces branch cut discontinuities as time to maturity increases, causing systematic mispricings that are invisible at short maturities but accumulate at longer maturities. Albrecher, Mayer, Schoutens & Tistaert (2007) call this problem "The Little Heston Trap" and show that two algebraically equivalent forms of the characteristic function circulate in literature, of which one is numerically unstable at longer maturities due to its choice of complex branch. Papers published since have consistently adopted the stable form, and it is this form that is implemented throughout this study:

$$\begin{aligned}
d &= \sqrt{(\rho\sigma iu - \kappa)^2 + \sigma^2(iu + u^2)} \\
g &= \frac{\kappa - \rho\sigma iu - d}{\kappa - \rho\sigma iu + d} \\
D &= \frac{\kappa - \rho\sigma iu - d}{\sigma^2} \frac{1 - \exp(-d\tau)}{1 - g \exp(-d\tau)} \\
C &= \frac{\kappa\theta}{\sigma^2} \left[ (\kappa - \rho\sigma iu - d)\tau - 2 \ln \frac{1 - g \exp(-d\tau)}{1 - g} \right]
\end{aligned}$$

where  $\tau = T - t$ . The characteristic function is recovered as

$$\phi(u; \tau) = \exp(iu(\ln S_0 + (r - q)\tau) + C(u, \tau) + D(u, \tau) v_0) \quad (13)$$

where  $S_0$  is the initial asset price,  $v_0$  is the initial variance, and  $q$  is the dividend yield. Unlike the pair of index-dependent characteristic functions  $\phi_1, \phi_2$  required for Gil-Pelaez inversion (equation 2.1.3), the COS method requires only this single characteristic function of the log-asset price, evaluated once per candidate parameter.

Fang & Oosterlee (2008) propose the COS method as an alternative to Fourier-based inversion. Rather than using a quadrature-based method for the inversion like Gil-Pelaez, COS reconstructs the log-return density from a truncated cosine series on a finite interval  $[a, b]$ .

The derivation begins from the risk-neutral pricing formula:

$$V(x, t_0) = e^{-r\Delta t} \int_{-\infty}^{\infty} g(y, T) f(y|x) dy \quad (14)$$

where  $x = \ln(S_0/K)$ ,  $y = \ln(S_T/K)$ ,  $g(y, T)$  is the discounted payoff as a function of the log-asset price, and  $f(y|x)$  is the probability density. Since  $f$  is not available in closed form for Heston, COS approximates it on  $[a, b]$  using a truncated cosine series:

$$f(y|x) \approx \sum_{k=0}^{N-1} {}'A_k(x) \cos\left(k\pi \frac{y-a}{b-a}\right) \quad (15)$$

where  $\sum'$  denotes that the  $k = 0$  term is multiplied by  $\frac{1}{2}$ , and  $A_k(x)$  are the cosine coefficients of the density. The key is that these coefficients are identical to the Heston characteristic function  $\varphi_{hes}$  evaluated at a fixed set of points, so they follow directly from the closed-form expression without any numerical integration of the density itself. Specifically, if the function decays quickly, the density has negligible mass outside  $[a, b]$ , and so:

$$A_k(x) \approx \frac{2}{b-a} \operatorname{Re} \left[ \varphi_{hes} \left( \frac{k\pi}{b-a}; v_0 \right) \exp \left( -ik\pi \frac{a}{b-a} \right) \right] \quad (16)$$

For the Heston model, the characteristic function of the log-asset price is

$$\varphi_{hes}(\omega; v_0) := \exp(iu(r-q)\tau + C(\omega, \tau) + D(\omega, \tau) v_0) \quad (17)$$

Meaning  $\varphi_{hes}(\omega; v_0)$  is the closed-form expression of  $\varphi$  at  $x = 0$ , depending only on the Heston model parameters (including initial variance  $v_0$ ) rather than the current log-price. This factorisation allows  $e^{i\omega x}$  to be separated out and absorbed into the pricing sum, giving the Heston-specific COS pricing formula (Fang & Oosterlee 2008):

$$V(x, t_0, v_0) \approx K e^{-r\Delta t} \operatorname{Re} \left\{ \sum_{k=0}^{N-1} {}'\varphi_{hes} \left( \frac{k\pi}{b-a}; v_0 \right) U_k \exp \left( ik\pi \frac{x-a}{b-a} \right) \right\} \quad (18)$$

where for a European put the payoff coefficients are:

$$U_k = \frac{2}{b-a} (-\chi_k(a, 0) + \psi_k(a, 0)) \quad (19)$$

and  $\chi_k$ ,  $\psi_k$  are trigonometric integrals of the put payoff. The original payoff  $g(y, T)$  is absorbed analytically into the payoff coefficients  $U_k$ . In the general formulas of Fang & Oosterlee (2008),  $\chi_k$  and  $\psi_k$  are defined over a sub-interval  $[c, d] \subseteq [a, b]$ . However for a European put, the payoff is non-zero only for  $y \in [a, 0]$ , so substituting in  $c = a$  and  $d = 0$ :

$$\begin{aligned} \chi_k(a, 0) = \frac{1}{1 + \left(\frac{k\pi}{b-a}\right)^2} & \left[ \cos\left(k\pi \frac{-a}{b-a}\right) e^0 - \cos\left(k\pi \frac{0}{b-a}\right) e^a \right. \\ & \left. + \frac{k\pi}{b-a} \sin\left(k\pi \frac{-a}{b-a}\right) e^0 - \frac{k\pi}{b-a} \sin\left(k\pi \frac{0}{b-a}\right) e^a \right] \end{aligned} \quad (20)$$

$$\psi_k(a, 0) = \begin{cases} \frac{b-a}{k\pi} \left[ \sin\left(k\pi \frac{-a}{b-a}\right) - \sin(0) \right] & k \neq 0 \\ 0 - a & k = 0 \end{cases} \quad (21)$$

Applying the identities  $e^0 = 1$ ,  $\cos(0) = 1$ ,  $\sin(0) = 0$ ,  $\cos(-x) = \cos(x)$ , and  $\sin(-x) = -\sin(x)$  further simplifies them to:

$$\chi_k(a, 0) = \frac{1}{1 + \left(\frac{k\pi}{b-a}\right)^2} \left[ \cos\left(\frac{k\pi a}{b-a}\right) - e^a - \frac{k\pi}{b-a} \sin\left(\frac{k\pi a}{b-a}\right) \right] \quad (22)$$

$$\psi_k(a, 0) = \begin{cases} -\frac{b-a}{k\pi} \sin\left(\frac{k\pi a}{b-a}\right) & k \neq 0 \\ -a & k = 0 \end{cases} \quad (23)$$

The truncation interval  $[a, b]$  is determined from the cumulants (moments) of the log-return distribution. Beyond the fourth cumulant, despite a closed form being available, they become increasingly difficult to calculate and offer diminishing practical value.

Fang & Oosterlee (2008) recommend recovering call option prices via put-call parity rather than direct COS pricing of the call payoff to avoid sensitivity of the upper trun-

cation bound arising from the unbounded growth of the call payoff with the log-asset price.

Error enters COS pricing through domain truncation, series truncation at  $N$  terms, and approximation of density coefficients, each of which decays exponentially as  $N$  increases. Fang & Oosterlee (2008) demonstrate that  $N = 64$  terms achieves eight decimal digits of accuracy in under half a millisecond, an approximately twenty-fold speed improvement over the Carr-Madan FFT approach at equivalent accuracy, with negligible truncation error for maturities between 0.1 and 10 years. This combination of analytical exactness in the payoff coefficients and exponential convergence in the density approximation is what makes COS the method of choice for repeated pricing during calibration.

## 2.3 Calibration Methodology

### 2.3.1 Calibration Approach and Loss Function

The Heston model can be calibrated either by fitting a time series of prices for a fixed contract or by fitting the implied volatility surface across multiple strikes and maturities for a fixed date (Cui et al. 2017). The latter is what is usually used in real-time pricing and risk management. Given observed market implied volatilities  $\sigma_i^{\text{mkt}}$  for a set of  $N$  options indexed by strike and maturity, calibration minimises a loss function  $L(\Theta)$  over the parameters  $\Theta = (\kappa, \theta, \sigma, \rho, v_0)$ .

Christoffersen & Jacobs (2004) demonstrated that the choice of  $L$  has material consequences on the success of the model. The study shows empirically that calibrating to minimise implied volatility root mean squared error (IV RMSE) specifically produces superior out-of-sample performance relative to loss functions specified in terms of option prices.

$$L(\Theta) = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{\sigma}_i^{\text{IV}}(\Theta) - \sigma_i^{\text{mkt}})^2} \quad (24)$$

where  $\hat{\sigma}_i^{\text{IV}}(\Theta)$  is the model-implied volatility for option  $i$ . This is obtained by solving  $C_{\text{BSM}}(\hat{\sigma}_i^{\text{IV}}) = C_i^{\text{model}}(\Theta)$  numerically for each option  $i$ . They further show that the loss

function used during parameter estimation should be the same one used for subsequent evaluation: optimising one objective and assessing fit by another introduces model misspecification into estimation objective differences, making it impossible to isolate either effect.

Calibration error is not the only consideration, however. Guillaume & Schoutens (2012) describe calibration risk as the sensitivity of model prices to the particular parameter set obtained during calibration. This is particularly relevant for models such as Heston, where different parameter combinations may produce similar calibration errors but materially different prices. Calibration should therefore be considered not only in terms of goodness of fit, but also the stability and economic appropriateness of the resulting parameter estimates.

### 2.3.2 Optimisation Methods

Minimising the Heston calibration objective over the five-dimensional parameter space is a non-trivial numerical optimisation problem. Cui et al. (2017) note that there is no consensus in the literature on if the calibration objective is convex or not, and that parameter dependencies (like the tendency of  $\kappa$  and  $\sigma$  to offset each other in their joint effect on the variance process) can produce broad flat regions near the optimum in which different parameter combinations can produce similar loss values. Furthermore, Gilli & Schumann (2011) demonstrate that standard gradient-based methods often fail for Heston calibration, producing solutions that are extremely sensitive to the choice of starting values; that restarting from different initial conditions can yield very different parameter estimates even when the final minimised loss values are near-identical. These observations motivate considering a range of optimisation strategies. The literature groups approaches into three categories: local gradient-based methods such as Levenberg-Marquardt (Levenberg 1944, Marquardt 1963), gradient-free direct search like Nelder-Mead (Nelder & Mead 1965), and global population-based algorithms such as Differential Evolution (Storn & Price 1997). Hybrid approaches also exist: Gilli & Schumann (2011) augment Differential Evolution with a Nelder-Mead direct search component, and

simulated annealing (a technique that occasionally accepts a worse potential solution, with this probability decreasing over the course of the search, to allow escape from local minima) can be layered onto Nelder-Mead to allow the occasional uphill move. However, Storn & Price (1997) find even annealed Nelder-Mead insufficient for reliable minimisation.

Each type has a different trade-off between computational cost and resistance to local minima traps.

**Levenberg-Marquardt.** The Levenberg-Marquardt (LM) algorithm, introduced independently by Levenberg in 1944 and by Marquardt in 1963, is a gradient-based method for nonlinear least-squares problems. In each iteration, the loss function gets updated thusly:

$$\Delta\Theta = (J^\top J + \mu I)^{-1} \nabla L \quad (25)$$

where  $J$  is the Jacobian of the residual vector,  $I$  the identity matrix, and  $\mu$  a damping factor. When  $\mu$  is large, the  $\mu I$  term dominates  $J^\top J$  and the update reduces to a small gradient descent step. As  $\mu$  decreases toward zero,  $J^\top J$  dominates and the method becomes a Gauss-Newton step, using curvature information for fast convergence near the optimum. The algorithm adapts  $\mu$  automatically by increasing it when a step fails to reduce the loss and decreasing it when progress is good. The Hessian is approximated as  $J^\top J$  near the optimum when residuals are small. Its main limitation is susceptibility to local minima: following the gradient from a given starting point, the algorithm may converge to a local solution rather than the global one, particularly on non-convex surfaces. Cui et al. (2017) cite LM with numerical gradient as the fastest established method for Heston calibration.

**Nelder-Mead.** The Nelder-Mead (NM) simplex method is a gradient-free local optimiser. Rather than computing gradient information, NM keeps a geometric simplex of  $D + 1$  vertices in the  $D$ -dimensional parameter space (meaning six vertices for the five-parameter Heston calibration problem) and iteratively updates it through a sequence

of moves that shifts the simplex towards regions of lower objective value. The method can even aptly handle discontinuous objective functions - something which would break gradient-based methods. Gilli & Schumann (2011) apply NM to Heston calibration within a hybrid search framework. The main limitation they find is that the acceptance criterion of a vertex successfully minimising the objective is greedy such that the simplex is unable to escape local minima. Storn & Price (1997) note that introducing a simulated annealing element again offers only partial improvement.

**Differential Evolution.** Differential Evolution (DE) is a population-based global optimisation algorithm. Rather than maintaining a single candidate solution, DE evolves a population of  $N_P$  parameter vectors across successive generations. In each generation, a trial vector is constructed for each population member, introducing a mutation strategy that automatically adapts the step size to the current population spread. This gives DE both broad exploration in flat regions and quick convergence near the optimum. The trial vector is compared element-wise with the original, and replaced if the resulting objective value is lower. This population-based mechanism enables simultaneous exploration of many regions of the parameter space and escape from local minima that trap aforementioned method types. Storn & Price (1997) demonstrate that DE outperforms both simulated annealing and annealed Nelder-Mead on multiple benchmark functions in terms of required number of function evaluations. Gilli & Schumann (2011) apply DE to Heston calibration and confirm that good surface fits are achievable, while noting that convergence of individual parameter estimates proceeds more slowly than convergence of the loss objective. The population-based mechanism that gives DE its global search property requires substantially more function evaluations than local methods, demonstrating the computational cost of the global approach.

## 2.4 Calibration During Stressed Regimes

While the literature addresses the relative strengths and weaknesses of alternative Heston calibration methods, there appears to be a gap in studies directly comparing their

performance during sustained periods of elevated market volatility. This matters because a persistently high volatility can alter the behaviour of Heston's mean-reverting process and the interaction between its parameters relative to calmer conditions. This means that evidence of an optimiser's effectiveness under conventional market conditions does not necessarily generalise to sustained high-volatility periods. This dissertation addresses this gap by comparing the quality and stability of Heston calibration across alternative optimisation methods using option data from such conditions.

## 3 Methodology

### 3.1 Research Design Overview

This study compares three Heston calibration optimisers: Levenberg-Marquardt (LM), Nelder-Mead (NM), and Differential Evolution (DE), while keeping the pricing engine COS and the loss function implied volatility RMSE fixed. All three optimisers are applied to the same dataset, under two market regimes: calm (2021) and stressed (2022). For each optimiser and regime, calibration quality (IV RMSE), parameter stability (variation in  $\kappa, \theta, \sigma, \rho, v_0$  across calibration dates), computational cost (number of function evaluations), and out-of-sample pricing error are recorded and compared.

### 3.2 Data

Historical SPX daily close data is obtained from OptionDX. The calm period is defined as 1 January to 31 October 2021, and the stressed period as 1 January to 31 October 2022, spanning the S&P 500's decline from its all-time high on 3 January 2022 to its trough on 12 October 2022. This follows Liu, Jiang & Zhang (2025), who characterise 2021 as a period of persistent record highs and 2022 as having frequent, sustained declines in the index.

To keep the total number of calibrations computationally tractable, one calibration date per month is used, taken as the second Wednesday of each month, giving ten dates per regime. Where no quotes are available for the second Wednesday (as occurred on 11 May 2022), the preceding Tuesday is used instead.

Each date's option chain is filtered as follows:

1. Maturities are restricted to between approximately 2 months (specifically 0.166 years) and 5 years. This is well within the range over which Fang & Oosterlee (2008) demonstrate negligible COS truncation error, and a little more conservative than the 30 days used by Cui et al. (2017), given the short-maturity limitations of Heston discussed in Section 2.

2. Quotes with a zero trading volume, zero bid, or with bid exceeding ask are removed, to mitigate the possibility of stale and misleading quote data being included in the data set.
3. The risk-free rate  $r$  is set at the federal funds rate in place at the time of each quote date.

### 3.3 Pricing Implementation

The Heston characteristic function is implemented in the stable form as recommended by Albrecher et al. (2007). The COS inversion method of Fang & Oosterlee (2008) is used with  $N = 64$  terms and a truncation width parameter  $L = 12$  (as was used in the paper), and the truncation range is centred at  $x_0 + c_1$  as taken from Ruijter & Oosterlee (2012), where  $x_0 = \ln(S_0/K)$ , rather than at  $c_1$  alone as given in the original Fang & Oosterlee (2008). The second cumulant  $c_2$  used to determine this range follows the corrected formula of Le Floc'h (2020), who identify an error in the originally published expression in Fang & Oosterlee (2008).

Original:

$$\begin{aligned}
c_2 = \frac{1}{8\kappa^3} & \left( \kappa\theta T e^{-\kappa T} (v_0 - \theta)(8\kappa\rho - 4\kappa) + \kappa\rho\theta(1 - e^{-\kappa T})(16\theta - 8v_0) \right. \\
& + 2\theta\kappa T(-4\kappa\rho\theta + \sigma^2 + 4\kappa^2) + \sigma^2 \left( (\theta - 2v_0)e^{-2\kappa T} + \theta(6e^{-\kappa T} - 7) + 2v_0 \right) \\
& \left. + 8\kappa^2(v_0 - \theta)(1 - e^{-\kappa T}) \right) \quad (26)
\end{aligned}$$

Le'Floch:

$$\begin{aligned}
c_2 = \frac{v_0}{4\kappa^3} & \left\{ 4\kappa^2 [1 + (\rho\sigma t - 1)e^{-\kappa t}] + \kappa [4\rho\sigma(e^{-\kappa t} - 1) - 2\sigma^2 t e^{-\kappa t}] + \sigma^2(1 - e^{-2\kappa t}) \right\} \\
& + \frac{\theta}{8\kappa^3} \left\{ 8\kappa^3 t - 8\kappa^2 [1 + \rho\sigma t + (\rho\sigma t - 1)e^{-\kappa t}] \right. \\
& \left. + 2\kappa [(1 + 2e^{-\kappa t})\sigma^2 t + 8(1 - e^{-\kappa t})\rho\sigma] + \sigma^2(e^{-2\kappa t} + 4e^{-\kappa t} - 5) \right\} \quad (27)
\end{aligned}$$

Call prices are obtained from COS via put-call parity, following the recommendation

of Fang & Oosterlee (2008). The Feller condition is purposely not enforced as a calibration constraint, as recommended by Cui et al. (2017).

### 3.4 Loss Function

Calibration minimises implied volatility RMSE, as defined in equation (24), as recommended by Christoffersen & Jacobs (2004). Model implied volatilities are obtained by inverting the price derived with COS through the BSM formula using `scipy`'s Brent method. Levenberg-Marquardt's implementation in `scipy` requires that failed implied volatility values not be dropped as the vector needs to remain the same length, and so inversions are instead assigned a fixed penalty of 5.0. The search bracket used for inversion is bounded above by  $\sigma = 5.0$ , so this penalty exceeds any genuine residual that could otherwise occur. For NM and DE, which have no such requirement, failed inversions are instead dropped from the RMSE calculation.

### 3.5 Starting Parameters

Starting values follow the heuristics summarised by Cui et al. (2017).  $\kappa$  is set to  $2.75/\tau_{\min}$ , following Clark's heuristic, where  $\tau_{\min}$  is the shortest maturity present in the filtered chain of the given quote date.  $v_0$  and  $\theta$  are both set to the squared mean implied volatility of the ten options nearest to at-the-money. No heuristic could be found in the literature for  $\sigma$  or  $\rho$  and so they are set to 0.5 and  $-0.5$  respectively, which is consistent with calibrated equity index parameters reported elsewhere (e.g.  $\sigma \approx 0.41$ ,  $\rho \approx -0.52$  in Forde & Jacquier (2009)). Differential Evolution does not use starting values and instead initialises its population directly from the parameter bounds.

### 3.6 Optimiser Configuration

NM and DE share the parameter bounds  $\kappa \in [0.01, 20]$ ,  $\theta \in [0.001, 1]$ ,  $\sigma \in [0.01, 5]$ ,  $\rho \in [-0.99, 0.99]$ ,  $v_0 \in [0.001, 1]$ . `scipy`'s LM implementation does not support bounds. In pursuit of tolerable runtimes in this study, convergence tolerances were set somewhat loose after some trial and error and differ across optimisers: LM uses  $\text{xtol} = \text{ftol} = 0.01$ , NM

uses  $\text{xatol} = \text{fatol} = 0.07$  with a maximum of 1000 iterations, and DE uses a population-spread tolerance of 0.7, a population size of 8, and a maximum of 50 generations, with a fixed random seed for reproducibility. DE’s tolerance is substantially looser than LM and NM’s because it was found to require several hours per calibration at tighter settings without a proportionate gain in final accuracy. Differential Evolution was run without parallelisation, following instability observed when using `scipy`’s multiprocessing workers in the computing environment used for this study.

### 3.7 Validity Screening

Following calibration, the results were screened for a reported convergence that did not correspond to a genuinely reliable fit, including

- $\theta$ ,  $\sigma$ , or  $v_0 > 1$
- $|\rho| > 1$
- $\kappa$  or  $\sigma$  at or near their upper bounds
- termination via iteration limit rather than satisfaction of the set tolerance

No calibration among the 60 performed (three optimisers, two regimes, ten dates each) was flagged by any of these criteria.

### 3.8 Post-Analysis

Following calibration, five further analyses were carried out. First, parameter stability was assessed for each optimiser and regime, by computing the mean and standard deviation of  $\kappa$ ,  $\theta$ ,  $\sigma$ ,  $\rho$ , and  $v_0$  across the ten dates in each regime.

Out-of-sample performance was evaluated by selecting a small sample of representative dates  $t$  in each regime, and applying the parameters already calibrated to that date, without further optimisation, to the option surface observed five trading days later; at  $t + 5$ . The resulting model implied volatilities were compared to the market implied volatilities quoted on  $t + 5$  to obtain an out-of-sample IV RMSE, using the same pricing and inversion procedure used previously. This required only loading and cleaning the

$t + 5$  surface for each regime.

Computational cost was compared across the three optimisers using the number of function evaluations recorded for each calibration, providing a measure of the effort required by each method independent of hardware.

Finally, the resultant implied volatility surface for each optimiser was plotted for the same single date and compared visually against the corresponding market implied volatility surface, with another plot showing the difference between the two for each regime.

## 4 Results

Table 1 reports the mean final calibrated parameter values and resultant IV RMSE across the ten dates in each regime for each optimiser.

| Optimiser | Regime | $\bar{\kappa}$ | $\bar{\theta}$ | $\bar{\sigma}$ | $\bar{\rho}$ | $\bar{v}_0$ | $\overline{\text{IV RMSE}}$ |
|-----------|--------|----------------|----------------|----------------|--------------|-------------|-----------------------------|
| DE        | calm   | 4.114          | 0.072          | 1.577          | -0.767       | 0.035       | 0.0066                      |
| DE        | stress | 1.758          | 0.112          | 0.892          | -0.785       | 0.059       | 0.0105                      |
| LM        | calm   | 3.511          | 0.073          | 1.460          | -0.768       | 0.036       | 0.0066                      |
| LM        | stress | 1.944          | 0.086          | 0.909          | -0.784       | 0.060       | 0.0105                      |
| NM        | calm   | 2.262          | 0.086          | 0.974          | -0.775       | 0.035       | 0.0113                      |
| NM        | stress | 2.164          | 0.071          | 0.805          | -0.789       | 0.063       | 0.0122                      |

Table 1: Mean calibrated parameters and IV RMSE by optimiser and regime, across ten calibration dates per regime.

The most interesting result is how close LM and DE finish across both regimes. Despite LM and DE being so fundamentally different in their search methods, the two methods converge to nearly identical final parameters and IV RMSE in both calm and stress. In calm, both achieve a mean RMSE of 0.0066, and in stress both achieve 0.0105. This is particularly impressive given that DE receives no starting parameters, while LM begins from the Cui et al. (2017) heuristics. Their convergence to the same answer from very different starting conditions provides strong evidence that a well-defined optimum exists in both regimes.

NM produces economically plausible parameters in both regimes but with a higher mean RMSE (0.0113 in calm and 0.0122 in stress) which suggests that its simplex-based search method consistently finds a less accurate solution than LM and DE. Notably,  $\rho$  is negative across all optimisers and both regimes, lying between  $-0.767$  and  $-0.789$  on average, as is well documented in literature for equity markets.

The finishing parameters of all three optimisers move sensibly between regimes:  $\theta$  and  $v_0$  both increase on moving from calm to stress which reflects the higher variance in 2022. Meanwhile,  $\kappa$  decreases which shows a slower mean reversion under sustained market stress.  $\rho$  becomes slightly more negative across all three optimisers which is consistent

with the steeper implied volatility skew generally observed in stressed equity markets (Gatheral et al. 2018).  $\sigma$  exhibits much less of a trend. This may partly reflect the interdependence between  $\kappa$  and  $\sigma$  in Heston calibration as stated by Cui et al. (2017), wherein the two parameters are not independently distinguishable from option prices alone. Given  $\kappa$ 's starting value of  $2.75/\tau_{\min}$  meant it typically started around 15 in this study before ending below 5, this meant the optimiser always had to traverse a great distance for  $\kappa$ . However since  $\sigma$  started at 0.5, it was already closer to its final value and so had to move much less, which may explain its relative stability.

## 4.1 Parameter Stability

Table 2 reports the standard deviation of each calibrated parameter and IV RMSE across the ten dates in each regime.

| Optimiser | Regime | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|-----------|--------|----------|----------|----------|--------|-------|---------|
| DE        | calm   | 2.660    | 0.012    | 0.531    | 0.050  | 0.017 | 0.0016  |
| DE        | stress | 1.385    | 0.065    | 0.308    | 0.040  | 0.023 | 0.0025  |
| LM        | calm   | 1.134    | 0.011    | 0.251    | 0.046  | 0.016 | 0.0016  |
| LM        | stress | 1.307    | 0.018    | 0.301    | 0.039  | 0.023 | 0.0025  |
| NM        | calm   | 1.904    | 0.031    | 0.430    | 0.114  | 0.014 | 0.0052  |
| NM        | stress | 0.625    | 0.012    | 0.120    | 0.061  | 0.024 | 0.0026  |

Table 2: Parameter stability by optimiser and regime: standard deviation across ten calibration dates per regime.

LM is the most stable optimiser in the calm regime, with the smallest standard deviations across every parameter ( $\kappa$  1.134,  $\sigma$  0.251,  $\rho$  0.046) and RMSE (0.0016). In the stressed regime, LM's standard deviations are similar to DE, with both showing similar variations in parameters across dates. What is also noteworthy is that unlike NM and DE, LM's standard deviations are very consistent between calm and stress.

DE's standard deviations are consistently small across regimes.  $\rho$  is particularly stable throughout, with a standard deviation of only 0.050 in calm and 0.040 in stress.

NM exhibits the opposite, showing considerably less stability in calm ( $\kappa$  1.904,  $\sigma$  0.430,  $\rho$  0.114) than in stress ( $\kappa$  0.625,  $\sigma$  0.120,  $\rho$  0.061). The standard deviation of

NM’s calm IV RMSE (0.0052) is small compared to its mean (0.0113) which suggests NM’s fit quality was somewhat consistent across calm dates despite finishing with a less accurate solution for each date. NM behaving more erratically in the calm regime than in the stressed regime suggests that its simplex search is more sensitive to the landscape present for each calm date.

## 4.2 Computational Cost

Table 3 shows the number of function evaluations per calibration.

| Optimiser | Regime | Mean  | SD    | Min | Max |
|-----------|--------|-------|-------|-----|-----|
| DE        | calm   | 505.4 | 100.3 | 314 | 674 |
| DE        | stress | 504.2 | 129.4 | 308 | 782 |
| LM        | calm   | 13.6  | 2.8   | 8   | 17  |
| LM        | stress | 11.4  | 3.2   | 6   | 16  |
| NM        | calm   | 194.7 | 135.6 | 119 | 573 |
| NM        | stress | 173.2 | 75.5  | 96  | 325 |

Table 3: Computational cost by optimiser and regime: function evaluations per calibration across ten dates per regime.

LM requires far fewer evaluations than NM or DE, averaging 13.6 in calm and 11.4 in stress - around 45 times less than DE for almost identical results. This reflects LM’s use of the gradient to navigate directly toward the solution, whereas NM and DE rely only on function value comparisons.

DE’s evaluation count is very consistent across regimes (with a tight mean spread of 504 to 506) which reflects the population-based convergence method rather than sensitivity to the landscape. The standard deviations (100.3 calm, 129.4 stress) are quite small relative to the mean, suggesting DE’s convergence behaviour is stable irrespective of market conditions.

Computationally, NM sits between LM and DE, needing on average 194.7 evaluations in the calm regime and 173.2 in stressed. The higher count and larger standard deviation in calm (135.6 versus 75.5 in stress) again indicates NM varying more substantially across calm dates, with one date requiring 573 evaluations.

### 4.3 Out-of-Sample Performance

Tables 4, 5, and 6 report out-of-sample IV RMSE for three dates in early, mid, and late 2021 and 2022.

| Regime | Optimiser | In-Sample RMSE | OoS RMSE | $\Delta$ RMSE | $\Delta$ RMSE (%) |
|--------|-----------|----------------|----------|---------------|-------------------|
| calm   | LM        | 0.00802        | 0.01975  | +0.01174      | +146.4%           |
|        | NM        | 0.01319        | 0.02172  | +0.00854      | +64.7%            |
|        | DE        | 0.00802        | 0.01988  | +0.01186      | +147.9%           |
| stress | LM        | 0.00770        | 0.02582  | +0.01811      | +235.3%           |
|        | NM        | 0.01210        | 0.02546  | +0.01336      | +110.4%           |
|        | DE        | 0.00770        | 0.02582  | +0.01812      | +235.3%           |

Table 4: Out-of-sample IV RMSE, March: calm 2021-03-10  $\rightarrow$  2021-03-17; stress 2022-03-09  $\rightarrow$  2022-03-16.

| Regime | Optimiser | In-Sample RMSE | OoS RMSE | $\Delta$ RMSE | $\Delta$ RMSE (%) |
|--------|-----------|----------------|----------|---------------|-------------------|
| calm   | LM        | 0.00717        | 0.01821  | +0.01104      | +154.0%           |
|        | NM        | 0.01273        | 0.01972  | +0.00699      | +54.9%            |
|        | DE        | 0.00717        | 0.01828  | +0.01112      | +155.1%           |
| stress | LM        | 0.01690        | 0.02421  | +0.00732      | +43.3%            |
|        | NM        | 0.01706        | 0.02527  | +0.00821      | +48.1%            |
|        | DE        | 0.01690        | 0.02450  | +0.00760      | +45.0%            |

Table 5: Out-of-sample IV RMSE, May: calm 2021-05-12  $\rightarrow$  2021-05-19; stress 2022-05-10  $\rightarrow$  2022-05-17.

| Regime | Optimiser | In-Sample RMSE | OoS RMSE | $\Delta$ RMSE | $\Delta$ RMSE (%) |
|--------|-----------|----------------|----------|---------------|-------------------|
| calm   | LM        | 0.00489        | 0.02049  | +0.01560      | +319.2%           |
|        | NM        | 0.01781        | 0.03082  | +0.01301      | +73.1%            |
|        | DE        | 0.00489        | 0.02048  | +0.01559      | +319.1%           |
| stress | LM        | 0.01021        | 0.00828  | -0.00193      | -18.9%            |
|        | NM        | 0.01025        | 0.00812  | -0.00212      | -20.7%            |
|        | DE        | 0.01020        | 0.00821  | -0.00199      | -19.6%            |

Table 6: Out-of-sample IV RMSE, August: calm 2021-08-11  $\rightarrow$  2021-08-18; stress 2022-08-10  $\rightarrow$  2022-08-17.

In the calm regime, LM and DE degrade consistently and comparably across all three dates, with a  $\Delta\text{RMSE}$  ranging from +146% to +319%. The large figures moreso reflect the tight in-sample fits rather than poor out-of-sample performance, considering the raw degradation ranges only from 0.011 to 0.016. NM also degrades consistently across all three calm dates (+64.7%, +54.9%, +73.1%), wherein the values being smaller than those for LM and DE again likely reflect its higher in-sample RMSE fit rather than better out-of-sample flexibility.

In the stressed regime, degradation varies greatly across dates. May produces comparable degradation across all three optimisers (43-48%), while March is substantially higher (110-235%), which is consistent with March 2022 being a period of extreme market stress and volatility changes. August shows a slight improvement out-of-sample for all three optimisers, with finishing parameters describing the following week’s surface more closely than they described August 10th’s own surface. This indicates how difficult August 10th is to fit from scratch, whereas the following week does well with the nearby starting parameters.

#### 4.4 Implied Volatility Surfaces

The implied volatility surfaces for 13 October 2021 (calm) and 12 October 2022 (stress) are shown below for each optimiser. Each figure shows the market implied volatility surface (left), the model implied volatility surface (centre), and the error (difference) between model and market IV (right).

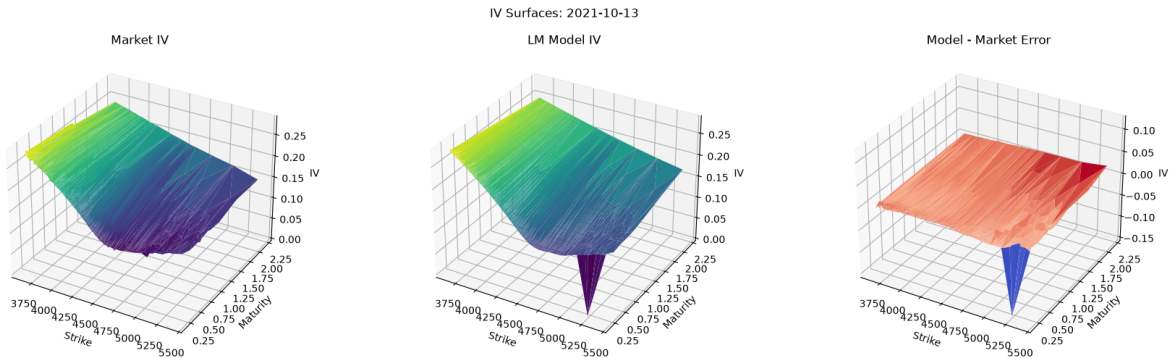


Figure 1: LM, calm regime, 13 October 2021.

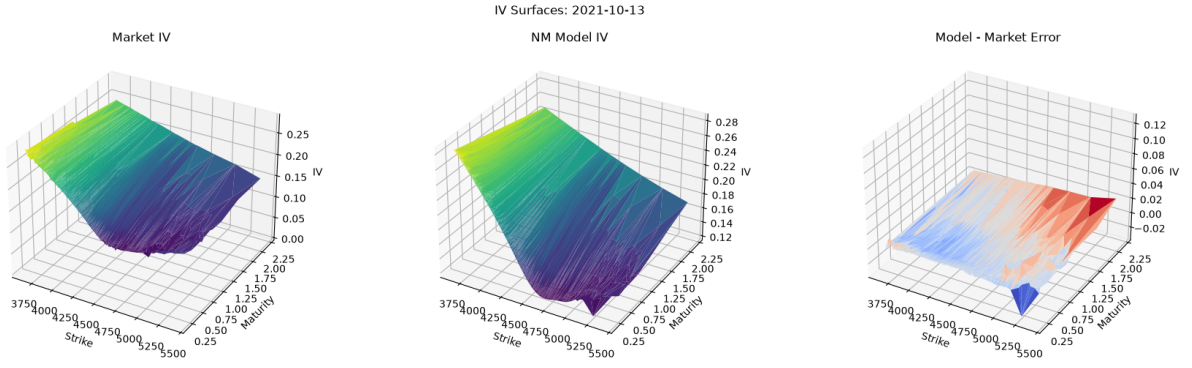


Figure 2: NM, calm regime, 13 October 2021.

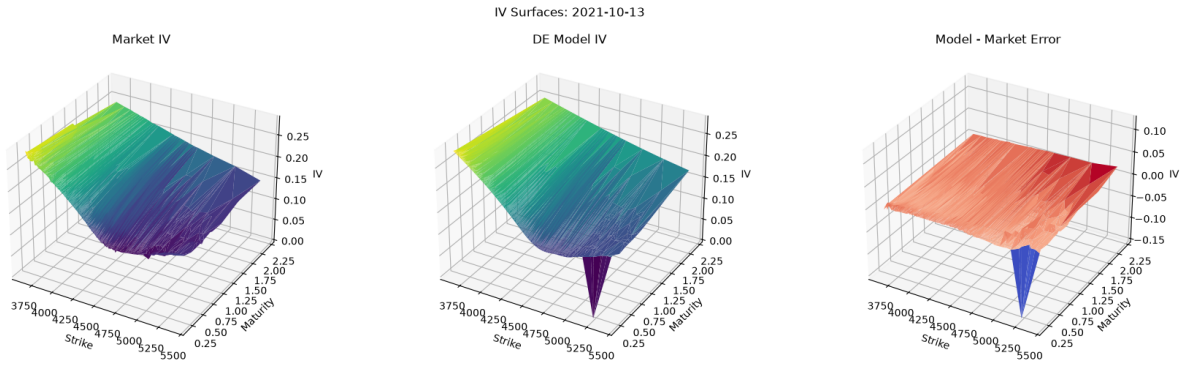


Figure 3: DE, calm regime, 13 October 2021.

In the calm regime (Figures 1-3), LM and DE produce visually indistinguishable fits to the market surface, with error surfaces that are nearly flat across most of the strike-maturity grid. Both show a spike at the shortest maturities, consistent with the known difficulty Heston has with fitting the short maturities. NM's fit visibly deteriorates noticeably at short tenors and in the far wings, suggesting the model struggles to capture the curvature in these regions.

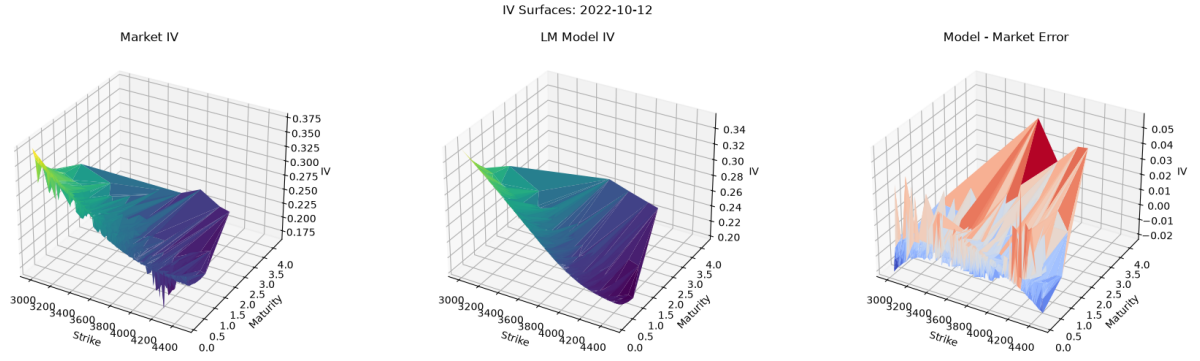


Figure 4: LM, stressed regime, 12 October 2022.

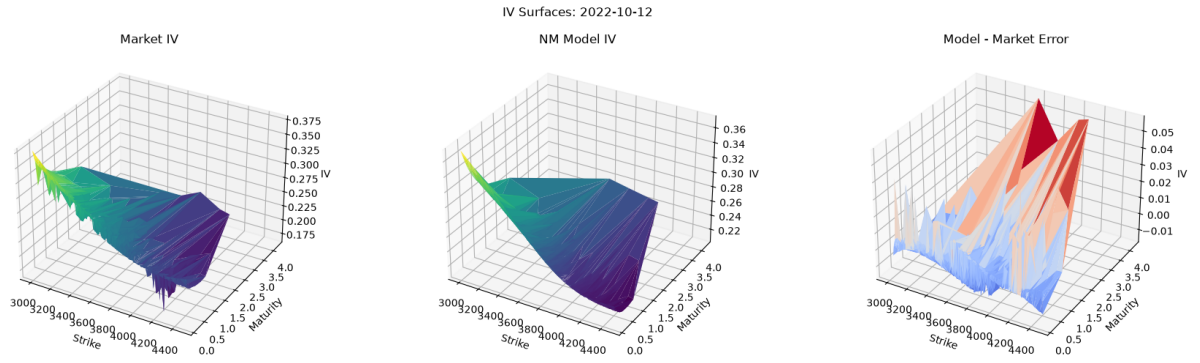


Figure 5: NM, stressed regime, 12 October 2022.

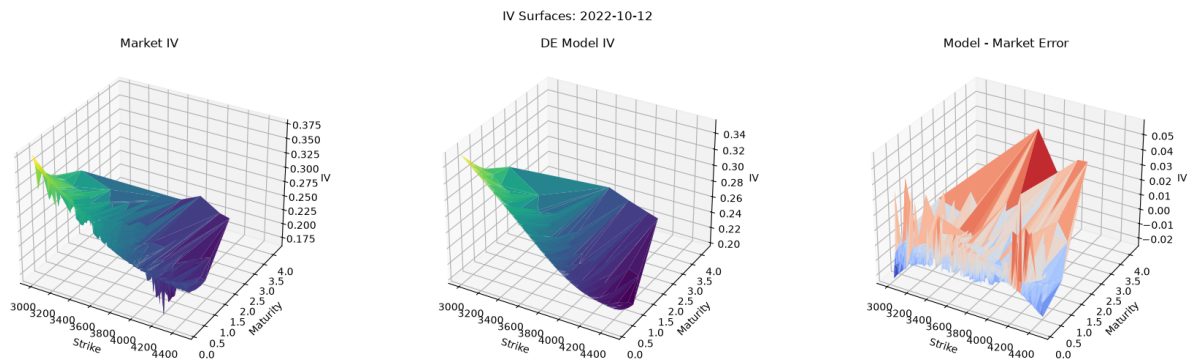


Figure 6: DE, stressed regime, 12 October 2022.

In the stressed regime (Figures 4-6), all three optimisers produce much more similar fits than in calm, with comparably jagged error surfaces at short maturities.

## 5 Discussion

### 5.1 Does Optimiser Choice Matter?

The most interesting finding of this study is the close agreement between LM and DE throughout. Despite using fundamentally different search mechanisms, the two methods converge to nearly identical parameters and IV RMSE in both calm and stress environments. That two such different approaches arrive at essentially the same answer strongly implies that a well-defined minimum exists in both regimes, and that the Heston loss surface (at least on data of this quality) is not so complex as to trap a local optimiser that has acceptable starting parameters. This is consistent with the literature suggesting that a lack of convexity matters less when the volatility surface is relatively smooth (Schoutens, Simons & Tistaert 2004), and this investigation finds that is still applicable to a stressed surface, where the surface is harder to fit but the optimal solution remains reachable.

NM's relative underperformance is the more informative finding on the question of whether optimiser choice matters. NM consistently produced a weaker in-sample fit than LM and DE in both regimes, with higher mean RMSE and, in calm, considerably higher variability across dates. Considering LM is known to be the most sensitive to starting parameters and NM used the same values, that cannot be deemed the root cause for how NM performed. The most likely explanation is simply that NM's simplex-based search is less effective at navigating the curvature of the Heston loss surface than either LM's Jacobian steps or DE's global population. On a smooth surface, a simplex can reliably reach the optimal solution (Nelder & Mead 1965), but on a more textured surface, it can struggle towards the wrong region. Overall, given sufficient and well-defined gradient information, LM is the cheapest and simplest of the three, matching DE's accuracy at a fraction of the computational cost.

## 5.2 Parameter Instability: Procedural or Structural?

Even the best performing optimisers will have a non-zero in-sample RMSE, and in this study, 0.007-0.010 remains in calm and a slightly higher 0.008-0.017 in stress. As described in section 2, Heston’s smooth variance dynamics are fundamentally unable to reproduce the rough volatility behaviour that has been documented in markets, and some lack of fit is therefore expected regardless of calibration method. This study provides a quantification of this across two distinct market regimes, and shows that smarter calibration narrows but does not eliminate it.

The parameter instability observed across dates is consistent with Cui et al. (2017)’s finding that many different parameter sets can produce similarly good fits to a given surface. Particularly, the aforementioned interdependence between  $\kappa$  and  $\sigma$  means that many combinations of the two parameters can produce similarly good fits to a given surface. The loss function therefore has a relatively flat section in this direction rather than a unique minimum, and the optimiser can terminate at different points along it depending on the starting conditions, producing parameter instability but no instability in fit quality. This is a structural property of the Heston model itself and is reflected in the similar instability patterns observed across LM and DE despite their very different search mechanisms.

## 5.3 Limitations

This study uses SPX options data only. SPX is the most liquid and studied equity index options market, which makes it an appropriate benchmark, but results may not generalise to less liquid assets, or non-equity derivatives, where the loss surface may be considerably noisier, more sensitive to stress, and the dependency on viable starting parameters may be more important.

The ”stressed” regime is represented by a single ten-month window in 2022. ”Stress” can present in different ways such as short-lived single-event crises, difficulties with liquidity, or sector-specific issues. These can have noticeably different volatility surfaces, and so these results may not generalise across all stress topologies. The findings should

therefore be understood as specific to the type of stress present throughout most of 2022 rather than as a general characterisation of stressed-regime calibration behaviour.

The range of ten dates per regime is sufficient to identify general patterns but limits the possibility of drawing stronger conclusions. The results reached here (especially the lack of trend in  $\sigma$ ) would likely benefit from a larger data sample.

Finally, COS requires specific parameter choices like setting  $L = 12$  and having  $N = 64$  terms. Alternative implementations, or the use of FFT-based pricing, may produce materially different loss surfaces and therefore different calibration outcomes, though broadly the outcomes are unlikely to be overly sensitive given their agreement with published benchmarks (Fang & Oosterlee 2008, Ruijter & Oosterlee 2012).

## 6 Conclusion

### 6.1 Practical Implications

The results of this study have direct implications for practitioners choosing a calibration methodology. Given that each evaluation involves pricing the full option surface using the COS method, this translates directly to runtime in production. LM is substantially faster for near-identical results. In a real-world trading or risk context, where calibration must be repeated across multiple assets, the efficiency LM exhibits for essentially identical results as the more complex DE gives it a clear advantage. This means that based only on accuracy of results, the case for using DE is not strong, as although it provides a robustness to starting point, this study demonstrates that a well-initialised LM calibration finds the same solution without that same robustness.

Nevertheless, sensitivity to starting parameters remains a practical issue for LM. When initialised far from the true optimum, LM can fail silently and report a convergence at a local minimum with no indication of the error beyond an elevated RMSE. A practitioner relying on LM alone would need some kind of systematic validation, be it a check of the RMSE against a threshold, or an occasional DE run as a validation step, to guard against this kind of failure. This kind of hybrid approach could protect against failure while keeping most of the benefits of LM's efficiency.

### 6.2 Future Work

There are many directions in which future work could be extended.

Alternative, more appropriate loss functions could be used. In the context of stress particularly, a vega-weighted RMSE can concentrate calibration on the at-the-money region where model fit would be most consequential to a practitioner.

The stress regime examined here represents one specific form of market stress. This study could be extended to other types to test how well the findings can be generalised across different abnormal volatility environments.

On the model side, the same calibration comparison framework could be applied to

the Rough Heston model instead, to test if a model known to better fit elevated volatility in addition to a wise calibrator choice could result in a better fit. Overall, it could be observed if optimiser choice interacts differently with a rougher loss surface.

A hybrid optimisation approach i.e. using DE to establish a baseline and LM to conduct the bulk of the search, could recover most of DE's robustness at a significantly lower computational cost than DE alone.

Finally, repeating the analysis on other derivative types would test how applicable the findings are beyond the highly liquid, well-studied SPX market to environments where the volatility surface is noisier and calibration is inherently more difficult.

## 7 Appendix

### 7.1 DE Raw Data

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2021-01-13 | 2.205    | 0.088    | 1.236    | -0.783 | 0.047 | 0.0057  |
| 2021-02-10 | 3.607    | 0.087    | 1.547    | -0.774 | 0.052 | 0.0088  |
| 2021-03-10 | 2.571    | 0.082    | 1.334    | -0.727 | 0.052 | 0.0080  |
| 2021-04-14 | 11.211   | 0.054    | 2.956    | -0.645 | 0.014 | 0.0091  |
| 2021-05-12 | 2.785    | 0.072    | 1.312    | -0.805 | 0.065 | 0.0072  |
| 2021-06-09 | 3.815    | 0.063    | 1.460    | -0.759 | 0.027 | 0.0051  |
| 2021-07-14 | 4.215    | 0.062    | 1.520    | -0.791 | 0.021 | 0.0048  |
| 2021-08-11 | 3.778    | 0.065    | 1.535    | -0.772 | 0.022 | 0.0049  |
| 2021-09-08 | 4.930    | 0.069    | 1.845    | -0.812 | 0.022 | 0.0060  |
| 2021-10-13 | 2.021    | 0.084    | 1.029    | -0.803 | 0.029 | 0.0068  |

Table 7: Per-date calibrated parameters and IV RMSE, DE, calm regime (2021).

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2022-01-12 | 4.149    | 0.069    | 1.479    | -0.767 | 0.024 | 0.0098  |
| 2022-02-09 | 3.047    | 0.069    | 1.136    | -0.808 | 0.039 | 0.0104  |
| 2022-03-09 | 1.309    | 0.097    | 1.013    | -0.803 | 0.085 | 0.0077  |
| 2022-04-13 | 3.461    | 0.077    | 1.248    | -0.809 | 0.044 | 0.0104  |
| 2022-05-10 | 1.563    | 0.075    | 0.720    | -0.865 | 0.089 | 0.0169  |
| 2022-06-08 | 0.808    | 0.096    | 0.611    | -0.786 | 0.058 | 0.0085  |
| 2022-07-13 | 0.217    | 0.260    | 0.621    | -0.769 | 0.063 | 0.0118  |
| 2022-08-10 | 2.035    | 0.078    | 0.790    | -0.750 | 0.042 | 0.0102  |
| 2022-09-14 | 0.798    | 0.101    | 0.631    | -0.771 | 0.061 | 0.0099  |
| 2022-10-12 | 0.188    | 0.201    | 0.675    | -0.718 | 0.088 | 0.0095  |

Table 8: Per-date calibrated parameters and IV RMSE, DE, stressed regime (2022).

## 7.2 LM Raw Data

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2021-01-13 | 2.204    | 0.088    | 1.235    | -0.783 | 0.047 | 0.0057  |
| 2021-02-10 | 3.464    | 0.087    | 1.516    | -0.774 | 0.052 | 0.0088  |
| 2021-03-10 | 2.564    | 0.082    | 1.334    | -0.727 | 0.052 | 0.0080  |
| 2021-04-14 | 5.372    | 0.060    | 1.816    | -0.657 | 0.022 | 0.0088  |
| 2021-05-12 | 2.739    | 0.072    | 1.305    | -0.804 | 0.065 | 0.0072  |
| 2021-06-09 | 3.823    | 0.062    | 1.460    | -0.759 | 0.027 | 0.0051  |
| 2021-07-14 | 4.205    | 0.062    | 1.518    | -0.791 | 0.021 | 0.0048  |
| 2021-08-11 | 3.791    | 0.065    | 1.537    | -0.772 | 0.022 | 0.0049  |
| 2021-09-08 | 4.930    | 0.069    | 1.847    | -0.811 | 0.022 | 0.0060  |
| 2021-10-13 | 2.018    | 0.084    | 1.028    | -0.803 | 0.029 | 0.0068  |

Table 9: Per-date calibrated parameters and IV RMSE, LM, calm regime (2021).

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2022-01-12 | 4.049    | 0.069    | 1.464    | -0.766 | 0.024 | 0.0098  |
| 2022-02-09 | 3.074    | 0.068    | 1.137    | -0.808 | 0.039 | 0.0104  |
| 2022-03-09 | 1.271    | 0.098    | 1.008    | -0.804 | 0.085 | 0.0077  |
| 2022-04-13 | 3.873    | 0.075    | 1.296    | -0.808 | 0.044 | 0.0104  |
| 2022-05-10 | 1.492    | 0.076    | 0.716    | -0.864 | 0.089 | 0.0169  |
| 2022-06-08 | 1.011    | 0.087    | 0.627    | -0.786 | 0.059 | 0.0086  |
| 2022-07-13 | 0.684    | 0.122    | 0.663    | -0.767 | 0.063 | 0.0119  |
| 2022-08-10 | 2.350    | 0.074    | 0.825    | -0.748 | 0.041 | 0.0102  |
| 2022-09-14 | 1.079    | 0.089    | 0.655    | -0.771 | 0.062 | 0.0099  |
| 2022-10-12 | 0.562    | 0.105    | 0.699    | -0.721 | 0.090 | 0.0095  |

Table 10: Per-date calibrated parameters and IV RMSE, LM, stressed regime (2022).

### 7.3 NM Raw Data

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2021-01-13 | 1.220    | 0.113    | 1.090    | -0.806 | 0.049 | 0.0066  |
| 2021-02-10 | 0.937    | 0.117    | 0.972    | -0.508 | 0.052 | 0.0185  |
| 2021-03-10 | 1.062    | 0.078    | 0.604    | -0.799 | 0.049 | 0.0132  |
| 2021-04-14 | 1.825    | 0.079    | 0.938    | -0.724 | 0.024 | 0.0105  |
| 2021-05-12 | 0.679    | 0.094    | 0.624    | -0.877 | 0.053 | 0.0127  |
| 2021-06-09 | 6.746    | 0.055    | 2.042    | -0.737 | 0.027 | 0.0054  |
| 2021-07-14 | 4.079    | 0.046    | 0.671    | -0.950 | 0.019 | 0.0165  |
| 2021-08-11 | 2.981    | 0.050    | 0.610    | -0.781 | 0.018 | 0.0178  |
| 2021-09-08 | 0.956    | 0.143    | 1.102    | -0.787 | 0.026 | 0.0065  |
| 2021-10-13 | 2.130    | 0.083    | 1.087    | -0.785 | 0.029 | 0.0057  |

Table 11: Per-date calibrated parameters and IV RMSE, NM, calm regime (2021).

| Date       | $\kappa$ | $\theta$ | $\sigma$ | $\rho$ | $v_0$ | IV RMSE |
|------------|----------|----------|----------|--------|-------|---------|
| 2022-01-12 | 1.978    | 0.057    | 0.651    | -0.762 | 0.032 | 0.0161  |
| 2022-02-09 | 3.259    | 0.059    | 0.903    | -0.846 | 0.040 | 0.0115  |
| 2022-03-09 | 2.182    | 0.059    | 0.759    | -0.862 | 0.086 | 0.0121  |
| 2022-04-13 | 3.167    | 0.063    | 0.918    | -0.825 | 0.048 | 0.0127  |
| 2022-05-10 | 2.488    | 0.069    | 0.817    | -0.866 | 0.096 | 0.0171  |
| 2022-06-08 | 1.458    | 0.072    | 0.639    | -0.803 | 0.061 | 0.0088  |
| 2022-07-13 | 1.982    | 0.089    | 0.934    | -0.710 | 0.066 | 0.0128  |
| 2022-08-10 | 1.630    | 0.081    | 0.716    | -0.763 | 0.042 | 0.0102  |
| 2022-09-14 | 1.678    | 0.078    | 0.751    | -0.749 | 0.064 | 0.0101  |
| 2022-10-12 | 1.817    | 0.086    | 0.966    | -0.698 | 0.098 | 0.0104  |

Table 12: Per-date calibrated parameters and IV RMSE, NM, stressed regime (2022).

## References

- Albrecher, H., Mayer, P., Schoutens, W. & Tistaert, J. (2007), ‘The little Heston trap’, *Wilmott Magazine* pp. 83–92.
- Bakshi, G., Cao, C. & Chen, Z. (1997), ‘Empirical performance of alternative option pricing models’, *The Journal of Finance* **52**(5), 2003–2049.
- Black, F., Scholes, M. & Merton, R. C. (1973), ‘The pricing of options and corporate liabilities’, *Journal of Political Economy* **81**(3), 637–654.
- Carr, P. & Madan, D. B. (1999), ‘Option valuation using the fast Fourier transform’, *Journal of Computational Finance* **2**(4), 61–73.
- Christoffersen, P., Heston, S. & Jacobs, K. (2009), ‘The shape and term structure of the index option smirk: Why multifactor stochastic volatility models work so well’, *Management Science* **55**(12), 1914–1932.
- Christoffersen, P. & Jacobs, K. (2004), ‘The importance of the loss function in option valuation’, *Journal of Financial Economics* **72**(2), 291–318.
- Cox, J. C., Ingersoll, J. E. & Ross, S. A. (1985), ‘A theory of the term structure of interest rates’, *Econometrica* **53**(2), 385–408.
- Cui, Y., del Baño Rollin, S. & Germano, G. (2017), ‘Full and fast calibration of the Heston stochastic volatility model’, *European Journal of Operational Research* **263**(2), 625–638.
- Fang, F. & Oosterlee, C. W. (2008), ‘A novel pricing method for European options based on Fourier-cosine series expansions’, *SIAM Journal on Scientific Computing* **31**(2), 826–848.
- Forde, M. & Jacquier, A. (2009), ‘Small-time asymptotics for implied volatility under the Heston model’, *International Journal of Theoretical and Applied Finance* **12**(6), 861–876.

- Gatheral, J., Jaisson, T. & Rosenbaum, M. (2018), ‘Volatility is rough’, *Quantitative Finance* **18**(6), 933–949.
- Gil-Pelaez, J. (1951), ‘Note on the inversion theorem’, *Biometrika* **38**(3/4), 481–482.
- Gilli, M. & Schumann, E. (2011), Calibrating option pricing models with heuristics, in A. Brabazon, M. O’Neill & D. Maringer, eds, ‘Natural Computing in Computational Finance’, Vol. 380 of *Studies in Computational Intelligence*, Springer, Berlin, Heidelberg, pp. 9–37.
- Guillaume, F. & Schoutens, W. (2012), ‘Calibration risk: Illustrating the impact of calibration risk under the Heston model’, *Review of Derivatives Research* **15**(1), 57–79.
- Heston, S. L. (1993), ‘A closed-form solution for options with stochastic volatility with applications to bond and currency options’, *The Review of Financial Studies* **6**(2), 327–343.
- Hull, J. & White, A. (1987), ‘The pricing of options on assets with stochastic volatilities’, *The Journal of Finance* **42**(2), 281–300.
- Le Floch, F. (2020), ‘More robust pricing of european options based on Fourier cosine series expansions’.
- URL:** <https://arxiv.org/abs/2005.13248>
- Levenberg, K. (1944), ‘A method for the solution of certain non-linear problems in least squares’, *Quarterly of Applied Mathematics* **2**(2), 164–168.
- Liu, Y., Jiang, P. & Zhang, G. (2025), ‘Stochastic volatility model with sticky drawdown and drawup processes: A deep learning approach’.
- URL:** <https://arxiv.org/abs/2503.14829>
- Lord, R. & Kahl, C. (2006), Optimal Fourier inversion in semi-analytical option pricing, Discussion Paper TI 2006-066/2, Tinbergen Institute. Available at SSRN: <https://ssrn.com/abstract=921336>.

- Marquardt, D. W. (1963), ‘An algorithm for least-squares estimation of nonlinear parameters’, *Journal of the Society for Industrial and Applied Mathematics* **11**(2), 431–441.
- Merton, R. C. (1976), ‘Option pricing when underlying stock returns are discontinuous’, *Journal of Financial Economics* **3**(1–2), 125–144.
- Nelder, J. A. & Mead, R. (1965), ‘A simplex method for function minimization’, *The Computer Journal* **7**(4), 308–313.
- Rubinstein, M. (1985), ‘Nonparametric tests of alternative option pricing models using all reported trades and quotes on the 30 most active CBOE option classes from august 23, 1976 through august 31, 1978’, *The Journal of Finance* **40**(2), 455–480.
- Ruijter, M. J. & Oosterlee, C. W. (2012), ‘Two-dimensional Fourier cosine series expansion method for pricing financial options’, *SIAM Journal on Scientific Computing* **34**(5), B642–B671.
- Schoutens, W., Simons, E. & Tistaert, J. (2004), ‘A perfect calibration! Now what?’, *Wilmott Magazine* pp. 66–78.
- Scott, L. O. (1987), ‘Option pricing when the variance changes randomly: Theory, estimation, and an application’, *Journal of Financial and Quantitative Analysis* **22**(4), 419–438.
- Stein, E. M. & Stein, J. C. (1991), ‘Stock price distributions with stochastic volatility: An analytic approach’, *The Review of Financial Studies* **4**(4), 727–752.
- Storn, R. & Price, K. (1997), ‘Differential evolution – A simple and efficient heuristic for global optimization over continuous spaces’, *Journal of Global Optimization* **11**(4), 341–359.
- Wiggins, J. B. (1987), ‘Option values under stochastic volatility: Theory and empirical estimates’, *Journal of Financial Economics* **19**(2), 351–372.